The empirical recurrence for OEIS A233887: recovering a conjecture that is not written down

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Abstract

OEIS A233887 records "Empirical recurrence of order 58", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 2401$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 58, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 58 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A233887 is "Number of (n+1)X(3+1) 0..6 arrays with every 2X2 subblock having the sum of the squares of the edge differences equal to 30, and no two adjacent values equal.". It has offset 1 and begins

$$1264, 9376, 65280, 496200, 3580624, 27131156, 197246776, 1489635452, \ \ldots$$

The entry states:

> Empirical recurrence of order 58 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 08:14:02, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2401$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S =$ 2401.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 4802$ exact terms of the model returns order 58 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{58} c_i\, a(n-i),$$

c_1	0	c_{16}	−379772848393	c_{31}	122171542643566	c_{46}	122437844456188
c_2	176	c_{17}	−124230108516	c_{32}	−3408946641617390	c_{47}	−8908307446448
c_3	32	c_{18}	2500153838987	c_{33}	−85976924413582	c_{48}	−26523105298144
c_4	−13110	c_{19}	717416233358	c_{34}	4197345979987647	c_{49}	1956572542208
c_5	−4024	c_{20}	−13138200066985	c_{35}	−143079081596	c_{50}	4059044588480
c_6	557085	c_{21}	−3217511387526	c_{36}	−4181913275202405	c_{51}	−295835011584
c_7	209642	c_{22}	55600736233816	c_{37}	89504281501640	c_{52}	−412012059392
c_8	−15389528	c_{23}	11272152535622	c_{38}	3348798045870528	c_{53}	29208473600
c_9	−6242702	c_{24}	−190689845225734	c_{39}	−130220796535786	c_{54}	24842804224
c_{10}	297665516	c_{25}	−30846887003066	c_{40}	−2135570432721643	c_{55}	−1700495360
c_{11}	121346888	c_{26}	532204323014373	c_{41}	112246170741038	c_{56}	−695181312
c_{12}	−4226362136	c_{27}	65431668963814	c_{42}	1071360744943143	c_{57}	44564480
c_{13}	−1653985546	c_{28}	−1211515514682464	c_{43}	−67046952552504	c_{58}	2621440
c_{14}	45490985291	c_{29}	−105351016011922	c_{44}	−416063638012388		
c_{15}	16513425642	c_{30}	2250779866841861	c_{45}	28789207285808		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 58, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $58+4$ terms removed returns order 58 as well, so $\deg q_{\min} = 58 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $58+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 58 that this sequence satisfies — and that family turns out to have exactly one member.

References

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